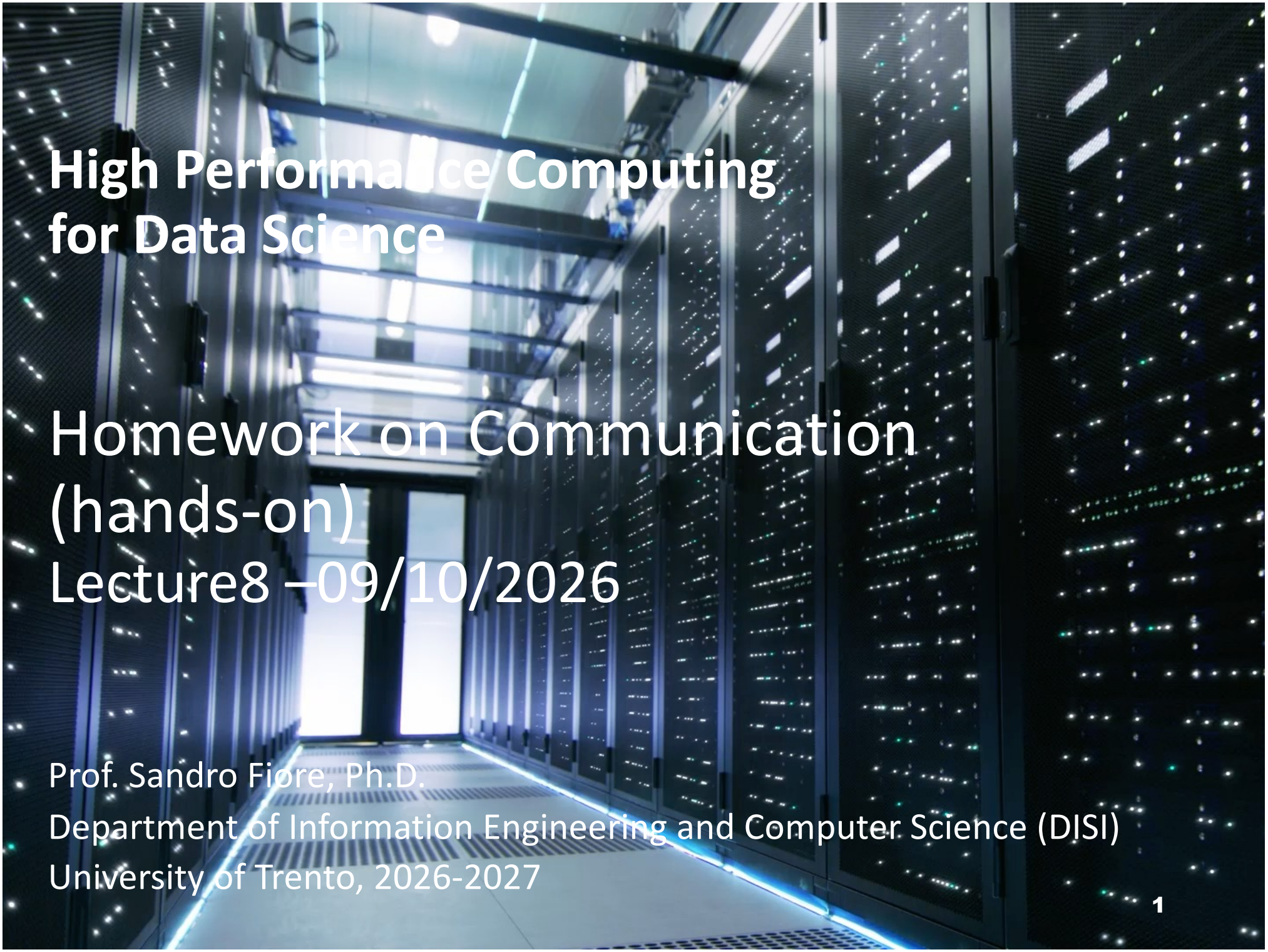


High Performance Computing for Data Science

Homework on Communication (hands-on)
Lecture8 –09/10/2026

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Department of Information Engineering and Computer Science (DISI)
University of Trento, 2026-2027

MPI example (from previous lectures)

```
1  #include <stdio.h>
2  #include <string.h>  /* For strlen             */
3  #include <mpi.h>     /* For MPI functions, etc */
4
5  const int MAX_STRING = 100;
6
7  int main(void) {
8     char       greeting[MAX_STRING];
9     int        comm_sz;  /* Number of processes */
10    int        my_rank;  /* My process rank     */
11
12    MPI_Init(NULL, NULL);
13    MPI_Comm_size(MPI_COMM_WORLD, &comm_sz);
14    MPI_Comm_rank(MPI_COMM_WORLD, &my_rank);
15
16    if (my_rank != 0) {
17       sprintf(greeting, "Greetings from process %d of %d!",
18             my_rank, comm_sz);
19       MPI_Send(greeting, strlen(greeting)+1, MPI_CHAR, 0, 0,
20             MPI_COMM_WORLD);
21    } else {
22       printf("Greetings from process %d of %d!\n", my_rank, comm_sz);
23       for (int q = 1; q < comm_sz; q++) {
24          MPI_Recv(greeting, MAX_STRING, MPI_CHAR, q,
25             0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
26          printf("%s\n", greeting);
27       }
28    }
29
30    MPI_Finalize();
31    return 0;
32 }  /* main */
```

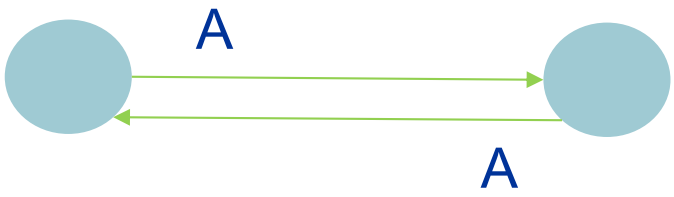


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Ping pong

Implement a "ping pong" using send & receive

Two processes one send and one receive each



Run it using: mpirun –n 2

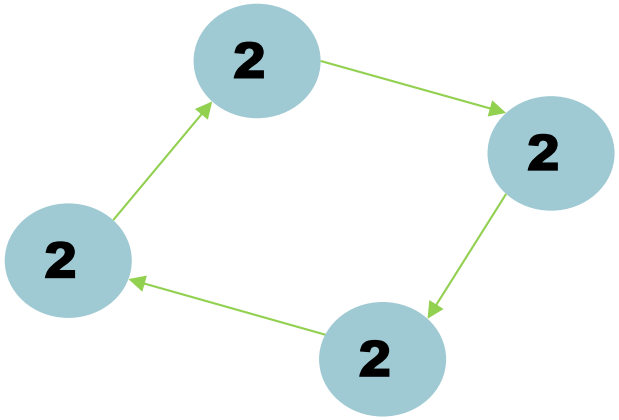
What can we argue about using n>2?

Would your application be resilient to that?

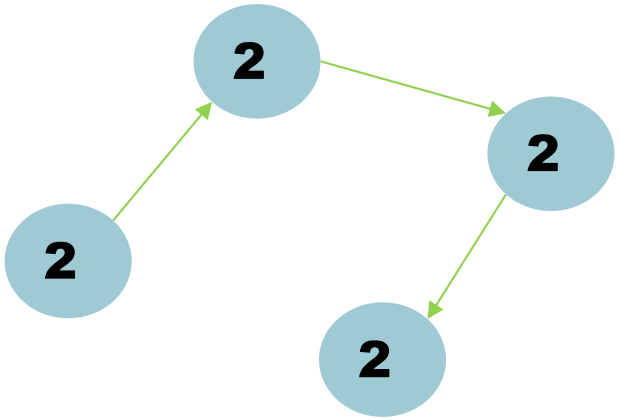
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Open and Closed Ring

Implement a ring by using send and receives



Option 1

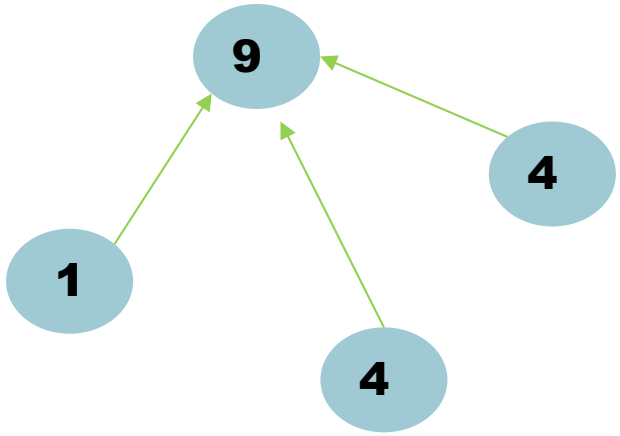


Option 2

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Reduce

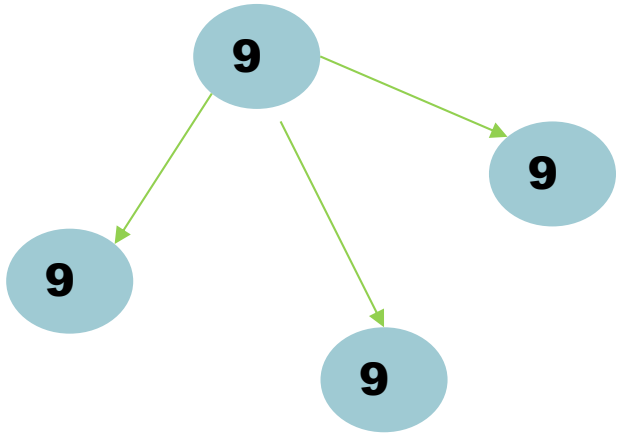
Implement a “reduce” by using send and receive + sum over the set of data collected



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Broadcast (loop-based)

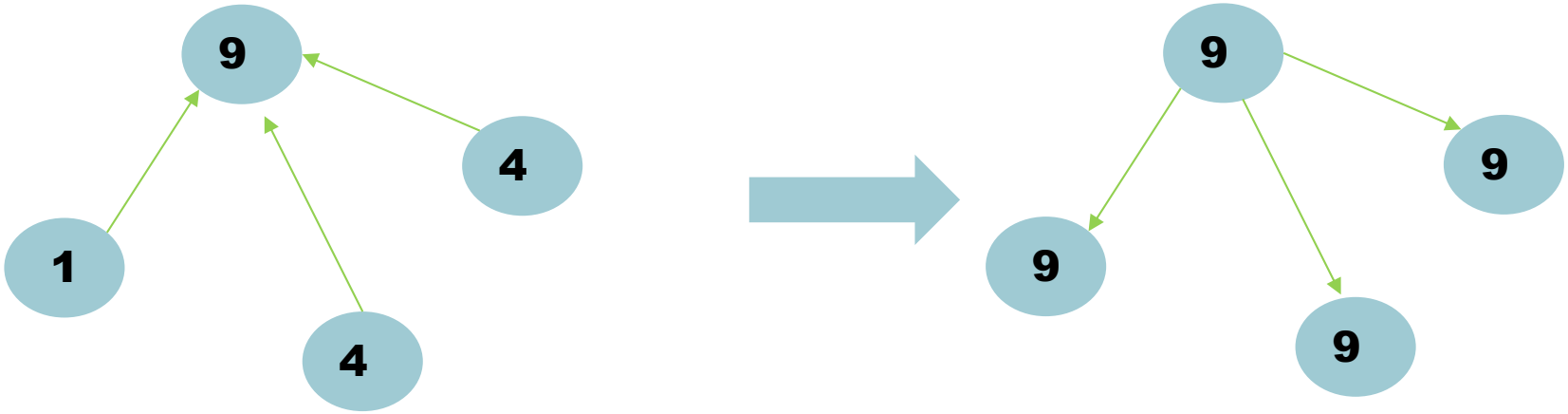
Implement a “broadcast” by using send and receive



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AllReduce

1) Implement a “reduce” by using send and receive + sum (or avg, min, max) over the set of data collected



2) …and then your “Allreduce”

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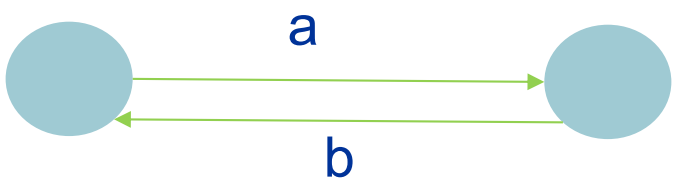
Ping pong (advanced)

Implement a "ping pong" using send & receive

Define the rules of the game!

One or more iterations? what data to exchange? exit condition?

Visual representation*?

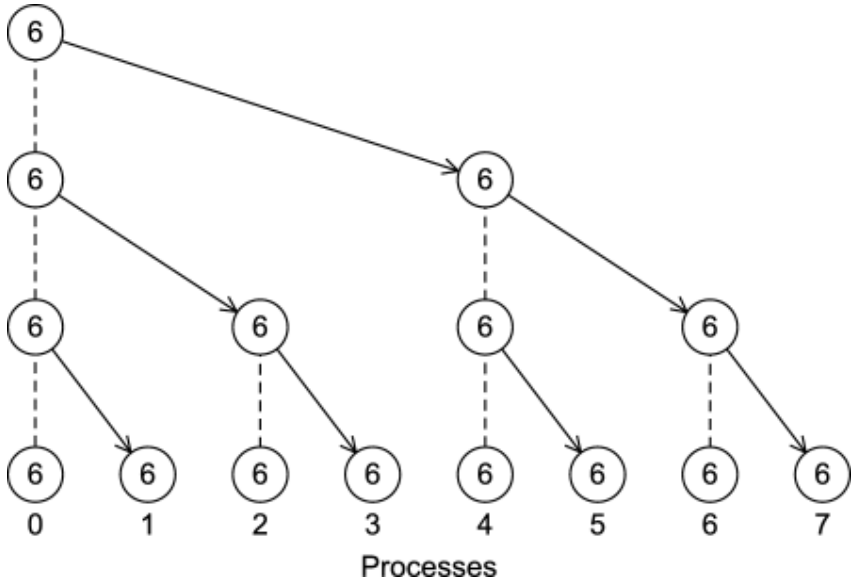


* Atari game "Pong" 1972

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Broadcast (tree-structured)

Implement a “broadcast” by using send and receive



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Feedback form

- Throughout the entire HPC4DS course there will be a form available for your feedback
  - Please provide any comment about:
    - Pros
    - Cons
    - Aspects that were not clear enough during the class
    - Any other feedback you think can be relevant for the course
    - …
- Information are gathered in an anonymous way

https://docs.google.com/forms/d/e/1FAIpQLScSVpNPfMP8poNNz-nLl9xlKlHkeBM2x_JgWyA0qQu6mOBl5A/viewform